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Joseph Henry Briggs

Day Month Year

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Chapter 1

Light and Gaussian Modes

This chapter will show the maths...

- check normalisation constants
- check the formula for the electric fields are actually right!
- change formulas to be in terms of a mode function and something that is to do with its propogational along z, and amplitude.

1.1 Plane Waves

Light is electromagnetic radiation. From Maxwell's equations it can be shown that the electric field, $\mathbf{E}$, component of electromagnetic radiation satisfies the wave equation

$$\nabla^2 \mathbf{E} = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \mathbf{E}. \quad (1.1)$$

A monochromatic, coherent, plane wave solution for $\mathbf{E}$ satisfies equation (1.1):

$$\mathbf{E} = \hat{\mathbf{n}}E \exp(i(\omega t - \mathbf{k} \cdot \mathbf{r} + \varphi)), \quad (1.2)$$

where $\hat{\mathbf{u}}$ indicates the polarisation, E is the complex field amplitude, ω is the angular frequency, $\mathbf{k}$ is the wave vector and φ is the absolute phase. The physical amplitude of the electric field oscillation is obtained from the $\Re(E)$. For optical systems is useful to define an optical axis. When the light is travelling along the optical axis, the $\mathbf{k} \cdot \mathbf{r}$ becomes kz . k is known as the wave number and is related to the wavelength of the light, λ by

$$k = \frac{2\pi}{\lambda}. \quad (1.3)$$

For plane waves there is no dependence of $\mathbf{E}$ on the x and y coordinates. The absolute phase φ becomes relevant in calculations involving interfering beams. By inserting equation (1.2) into equation (1.1), we obtain the dispersion relationship

$$\omega = ck. \quad (1.4)$$

The plane wave solution can be used to analyse many optical phenomena, such as amplitude modulation, phase modulation and interference.

1.2 Gaussian Modes

Light in an optical set-up is normally generated by a laser. Lasers produce almost monochromatic, coherent, light so in this sense they are quite similar to a plane wave, however lasers also have geometric properties. A laser produces light with an

approximately circular beam profile which is brightest at its centre. The spread of light power in a laser beam follows a Gaussian distribution, and the width of the beam profile is characterised by the radius at which $\sim 86\%$ of the light power is contained.

As the beam propagates along its optical axis, the beam size changes. A Gaussian beam has a point where it is smallest, this is known as the waist of the beam. A Gaussian beam can be completely characterised by the size of its waist and location along the waist along the optical axis. Whenever a Gaussian mode interacts with a simple optical component such as a lens or a mirror, the current Gaussian mode is transformed into a new Gaussian mode.

The following sections describe Gaussian modes and how they interact with real optical layouts with imperfections. The lowest order Gaussian mode will be derived, and the properties of this mode will be explained. The concept of the complex Gaussian beam parameter will be introduced as a way of encapsulating all the information needed to determine the properties of a Gaussian mode. Astigmatic Gaussian beams will be defined in terms of two complex Gaussian beam parameters. The first order Hermite-Gaussian (HG) mode will be shown to model the effect of misalignment, and then it will be described how all laser beams can be made up of a set of HG modes. Finally, the way that a Gaussian mode traverses an optical set-up will be expressed in terms of the ABCD matrix formalism.

1.3 The Paraxial Wave Equation

Short intro on what this is

To derive the paraxial wave equation, we apply the separation of variables to equation (1.1):

$$\mathbf{E}(\mathbf{r}, t) = E(\mathbf{r})T(t). \quad (1.5)$$

Next we reform equation 1.1 in terms of E and T

$$\frac{\nabla^2 E}{E} = \frac{1}{c^2 T} \frac{\partial^2 T}{\partial T^2}. \quad (1.6)$$

By noticing that either side of the equation (1.6) depends on $\mathbf{r}$ or t , the left hand side of this equation must equal a constant, and the Helmholtz equation can be obtained:

$$\frac{\nabla^2 E}{E} = -k^2. \quad (1.7)$$

Rewriting $E(\mathbf{r})$ in terms of two functions, **some how remove units from $u(\mathbf{r})$**

$$E(\mathbf{r}) = u(\mathbf{r}) \exp(-ikz), \quad (1.8)$$

and substituting this into equation (1.7) yields

$$\left(\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) u(\mathbf{r}) - 2ik \frac{\partial u}{\partial z} + \frac{\partial^2 u}{\partial z^2} \right) \exp(ikz) = 0. \quad (1.9)$$

The paraxial approximation means the $\frac{\partial^2 u}{\partial z^2}$ is neglected as it is much smaller than the other terms. Applying this approximation produces the paraxial Helmholtz equation

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) u(\mathbf{r}) - 2ik \frac{\partial u}{\partial z} = 0. \quad (1.10)$$

The physical interpretation of $u(\mathbf{r})$ is a function that describes the spatial distribution of the complex amplitude of the light field. **reword this so it actually makes sense.**

1.4 The fundamental Gaussian mode

Rewrite E in terms of u .

The simplest geometry of a laser beam can be described by the fundamental Gaussian mode. This is a mode which has a circular intensity profile that follows a Gaussian distribution, and the surface of constant phase is curved. Substituting the simplest solution to (1.10) into equation (1.8) produces the fundamental Gaussian mode **find out what the amplitude constant is**

$$E(\mathbf{r}) = E_0 \sqrt{\frac{2}{\pi}} \frac{1}{w(z)} \exp(i\phi(z)) \exp\left(-ik \frac{x^2 + y^2}{2R_C(z)} - \frac{x^2 + y^2}{w^2(z)}\right) \exp(-ikz). \quad (1.11)$$

This is characterised by a function $w(z)$, which gives the size of the beam at position z the optical axis, by $\phi(z)$, which gives the Gouy Phase of the beam at z along the optical axis, and by $R_c(z)$, which gives radius of curvature of the beam at z . The minimum value of $w(z)$ is known as the beam waist, or w_0 .

It is useful to express $w(z), \phi(z)$ and $R_c(z)$ in terms of the Rayleigh range z_R . The Rayleigh range characterises how fast the beam will spread out. A beam with a short Rayleigh length will expand quicker than a beam with a longer Rayleigh length. For example, in order for the beam to not be significantly clipped by the aperture of the test masses, the Rayleigh length of the Gaussian beam in the arm cavities at LIGO

has to be on the order of kilometres. The size of the waist determines the Rayleigh length, a smaller beam waist corresponds to a shorter Rayleigh length. If the waist of a Gaussian beam at $z = 0$, the position along the z axis by which the radius has increased by a factor of $\sqrt{2}$ is known as the Rayleigh range z_R , and is given by

$$z_R = \frac{\pi w_0^2}{\lambda}. \quad (1.12)$$

The radius of the beam as it propagates along the optical axis can be written as

$$w(z) = w_0 \sqrt{1 + \left(\frac{z}{z_R} \right)^2}. \quad (1.13)$$

The Rayleigh range can be used to approximately split beam into two regions: the near field for $z < z_R$ and the far field for $z > z_R$. In the far field equation (1.13) becomes linear and the beam size can be approximated as

$$w(z) \approx w_0 \frac{z}{z_R}. \quad (1.14)$$

As a Gaussian beam travels propagates along the optical axis, it acquires an additional phase shift compared to a plane wave. This is known as the Gouy phase shift, $\phi(z)$, and it given by

$$\phi(z) = \tan^{-1} \frac{z}{z_R}. \quad (1.15)$$

At either side of it waist, the fundamental Gaussian mode will be shifted $\pm\pi/2$ relative to a plane wave. An alternative way of putting this is the fundamental Gaussian mode will gain a π change as it goes through from $-\infty$ to ∞ compared to a plane wave. Most of this phase change is gained within $\pm z_R$ of its waist. The

Gouy phase results in the the phase velocity at the waist being greater for a Gaussian beam than it is for plane waves.

The radius of curvature, $R_c(z)$, in terms of the Rayleigh range is

$$R = z + \frac{z_R^2}{z}. \quad (1.16)$$

For $z \ll z_R$ the radius of curvature can be approximated as infinite. As the beam approaches its Rayleigh length the radius of curvature decreases until it reaches its minimum at $z = z_R$. After this the radius of curvature begin to flatten out again, and for $z \gg z_R$ the radius of curvature can be approximated as z . The defocus, $S(z)$, of a beam is a more practical way of describing a beam as it is easier to compute the effects of a beam interacting with a focussing element of an optical layout in terms of the defocus than it is with the radius of curvature. The defocus is defined as the inverse of its radius of curvature

$$S(z) = \frac{1}{R(z)} \quad (1.17)$$

The fundamental Gaussian mode can be expressed in terms of a complex beam parameter, $q(z)$. This parameter is useful for calculating how a gaussian beam is transformed by an optical system. From this parameter, all the physical quantities related to a Gaussian beam can be computed. The complex beam parameter is defined as

$$q(z) = z + iz_R. \quad (1.18)$$

The inverse of equation 1.18 can be written in terms of the curvature and radius of the beam:

$$\frac{1}{q(z)} = \frac{1}{R(z)} - i \frac{\lambda}{\pi w^2(z)}. \quad (1.19)$$

Equation (1.11) can be re written in terms of this the complex beam parameter,

$$E(\mathbf{r}) = E_0 \sqrt{\frac{2}{\pi}} \frac{q_0}{w_0 q(z)} \exp(i\phi(z)) \exp\left(-ik \frac{x^2 + y^2}{2q(z)}\right) \exp(-ikz). \quad (1.20)$$

The radius of the beam, gouy phase, and the defocus at a point z along the optical axis can be calculated from $q(z)$ using

$$w(z) = \sqrt{\frac{\lambda}{-\pi \Im\left(\frac{1}{q(z)}\right)}}, \quad (1.21)$$

$$\phi(z) = \tan^{-1}\left(\frac{\Re(q(z))}{\Im(q(z))}\right), \quad (1.22)$$

$$S(z) = \Re\left(\frac{1}{q(z)}\right). \quad (1.23)$$

Mode matching integral

1.4.1 Astigmatism

If a beam has an elliptical rather than circular profile, then the beam is astigmatic. For simple astigmatism, a beam can be described with two complex Gaussian beam parameters, one for the yz plane and one for the xz plane. These are referred to as the tangential (xz) and the saggital (xy) planes. A gaussian beam with simple astigmatism will have a different waist size and location in the tangential and sagittal planes. This type of astigmatism arises due to gaussian beams passing through a

cylindrical lens or having a non-zero angle of incidence to a mirror.

An simple astigmatic gaussian beam can be split up into two orthogonal fields
(rewrite in terms of $u(\mathbf{r})$)

$$\begin{aligned}
E(\mathbf{r}) &= E(x, z)E(y, z) \\
&= E_0 \sqrt{\frac{2}{\pi}} \sqrt{\frac{q_{x0}}{w_{x0}q_x(z)}} \sqrt{\frac{q_{y0}}{w_{y0}q_y(z)}} \exp(i\phi_{Av}(z)) \\
&\quad \times \exp\left(-ik\frac{x^2}{2q_x(z)} - ik\frac{y^2}{2q_y(z)}\right) \exp(-ikz). \tag{1.24}
\end{aligned}$$

where $\exp(i\phi_{Av}(z))$ is the average of the Gouy Phase in the saggital and tangential planes. The equations for waist size, equation (1.21), Gouy phase, equation (1.22) and defocus, equation (1.23), remain the same. A beam with simple astigmatism can be though of as two independent beams, and their interaction with the optical layout can be calculated independently.

1.5 Gaussian Beams Traversing optical Layouts

As a Gaussian beam propagates through an optical layout, the beam parameter that describes changes. The effect of simple optical components can be modelled with ABCD matrices, and an entire layout can be modelled by multiplying a series of ABCD matrices together. Components that introduce astigmatism to the beam can be modelled as having separate ABCD matrices for the tangential and saggital planes. This formalism can also be used to compute cavity eigenmodes.

When a beam interacts with an optical component it is transformed from a mode described by q_1 to a new mode q_2 according to

$$q_2 = \frac{Aq_1 + B}{Cq_1 + D} \quad (1.25)$$

where $n_{1,2}$ are the refractive indices of the medium $q_{1,2}$ are in. The coefficients A, B, C, D are defined by a coefficient matrix $\mathbf{M}$,

$$\mathbf{M} = \begin{pmatrix} A & B \\ C & D \end{pmatrix}. \quad (1.26)$$

To compute the coefficient matrix for a series of elements numbered 0 to n , the matrices for each individual element should be multiplied together

$$\mathbf{M} = \mathbf{M}_n \mathbf{M}_{n-1} \dots \mathbf{M}_0. \quad (1.27)$$

The ABCD matrices for various components is summarised in table 1.1.

1.6 Higher Order Modes and optical imperfections

The fundamental gaussian mode describes the electric field of a beam in an ideal optical system. Imperfections in the optical layout, such as misalignment and mode-mismatch, results in a different mode shape. If two modes are misaligned slightly, one will see a mode profile that looks like two spots instead of one. A small mode-mismatch results in a ring shaped mode profile. These imperfections in the layout and the resulting beam shapes can be modelled as the optical layout scattering light

Optical Component	ABCD Matrix
space	$\begin{pmatrix} 1 & d \\ 0 & 1 \end{pmatrix}$
Mirror or lens (normal incidence)	$\begin{pmatrix} 1 & 0 \\ -S & 1 \end{pmatrix}$
Mirror (AOI = α , Sagittal Plane)	$\begin{pmatrix} 1 & 0 \\ -\frac{S}{\cos \alpha} & 1 \end{pmatrix}$
Mirror (AOI = α , Tangential Plane)	$\begin{pmatrix} 1 & 0 \\ -S \cos \alpha & 1 \end{pmatrix}$

Table 1.1: Table of the ABCD matrices for simple optical components.

into higher-order optical modes.

There are many solutions to equation 1.10. One complete, orthonormal basis set of functions that can be used to construct solutions to 1.10 are the Hermite-Gaussian (HG) modes. These modes have xy symmetry, and appear as 'square' shaped. These are useful for modelling misalignments, as this type of imperfection can be naturally broken into xy coordinates. Mode-mismatches are best described by the $r\theta$ coordinates, and so its natural to express these in terms of another orthonormal basis set known as the Laguerre-Gauss (LG) modes. However, if the beam is astigmatic then the mode no longer has the symmetry required in order to be modelled by LG modes.

Equation (1.10) can be solved using HG modes, which have the mathematical form of

$$u_{nm}(x, y, z) = u_n(x, z)u_m(y, z). \quad (1.28)$$

The functions $u_{n,m}$ have the form

$$u_n(x, z) = \left(\frac{2}{\pi}\right)^{1/4} \left(\frac{\exp(i(2n+1)\phi(z))}{2^n n! w(z)}\right)^{1/2} H_n\left(\frac{\sqrt{2}x}{w(z)}\right) \exp\left(-i\frac{kx^2}{2R_c(z)} - i\frac{x^2}{w^2(z)}\right). \quad (1.29)$$

where $H_n(x)$ a Hermite polynomial of the order n . By writing $u_{nm}(x, y, z)$ out explicitly,

$$u_{nm}(x, y, z) = (2^{n+m-1} n! m! \pi)^{1/2} \frac{1}{w(z)} \exp(i(n+m+1)\phi(z)) \\ H_n\left(\frac{\sqrt{2}x}{w(z)}\right) H_m\left(\frac{\sqrt{2}y}{w(z)}\right) \exp\left(-ik\frac{x^2+y^2}{2R_c(z)} - i\frac{x^2+y^2}{w^2(z)}\right), \quad (1.30)$$

we can see that the higher order mode solutions to equation 1.10 have the same waist size and radius of curvature to the fundamental mode, but different intensity distributions and additional Gouy phase. If n and m are set to zero, then equation 1.11 is recovered. These HG modes interact with optical elements in the same way as the fundamental mode.

The eigen mode of a cavity is determined by its length and the radius of curvature of its end mirrors. The Gaussian mode that is the eigen mode of the cavity will have the same radius of curvature at the end mirrors of the cavity as the mirrors themselves. As higher order modes share the same radius of curvature as the fundamental

mode, these modes can also be resonant within the cavity. What separates these modes out in a cavity is the round trip Gouy phase that is acquired. Most cavities are designed such that the round trip Gouy phase is such that only one mode can be resonant at once, and all the other modes destructively interfere.

The Gouy phase separation of higher order modes is also important when designing alignment and mode matching control systems, as different types of errors have different phases. For example, angular and translational misalignment are both first order effects. This means they both transfer light from the HG_{00} mode into the HG_{10} . However, the complex amplitude of these misalignment situations is separated by $\pi/2$ rad of phase. Having two sensors separated by $\pi/2$ of Gouy phase, one can derive alignment control signals.

In general, a mode can be written as a superposition of HG modes

$$E(x, y, z) = \sum_{n=0}^{n=\infty} \sum_{m=0}^{m=\infty} A_m u(x) A_n u(y), \quad (1.31)$$

where A is the mode amplitude. This form is useful when calculating the coupling between modes.

The following example of the amount of power lost due to two arms of an interferometer being misaligned. The field at the AS port of the interferometer can be written as

$$E_{AS} = \frac{A_0}{\sqrt{2}} \left(\frac{\Delta A_0}{A_{00}} u_0 + \sum_{m=1}^{\infty} \frac{a_m u_m}{A_{00}} \right), \quad (1.32)$$

where A_0 is the incident field amplitude and A_{00} is the amplitude of the field in the arms. The change in the fundamental power due to an optical imperfection in one

of the arms is ΔA_0 . E_{AS} is normalised to the incoming power at the beam splitter. As the modes are orthogonal by calculating the power in the higher order modes, the power lost from the fundamental mode can be calculated. The amplitude of a mode can be written as

$$A_m = \frac{1}{\Gamma(m/2 + 1)2^m\sqrt{\pi}} \int_{-\infty}^{\infty} E(X)u_m(X)e^{-X^2/2}dX \quad (1.33)$$

where Γ is the Gamma function, X is the normalised distance $\sqrt{2}x/w$ and $E(X)$ is an arbitrary field distribution. For a misaligned beam with a purely angular misalignment θ , a linear phase gradient $\varphi(x) = 2\pi\theta x/\lambda$ must be applied to the regular field distribution. The field distribution at the waist can be written as

$$E(X) = \exp\left(iX\frac{\theta}{\theta_c} - \frac{X^2}{2}\right) \quad (1.34)$$

where the critical angle is $\theta_c = \lambda\sqrt{2}/\pi w$. Inserting this $E(X)$ into equation (1.33) gives

$$A_m = \frac{e^{-(\theta/\theta_c)^2}}{\Gamma(m/2 + 1)2^m\sqrt{\pi}} \int_{-\infty}^{\infty} e^{-(-X+i\frac{\theta}{\theta_c})^2}u_m(X)dX, \quad (1.35)$$

which can be solved analytically by using the relationship

$$\int_{-\infty}^{\infty} e^{-(z^*t)^2}u_n(t)dt = 2^n\sqrt{\pi}z^n \quad (1.36)$$

to give

$$A_m = \frac{(i\frac{\theta}{\theta_c})^m e^{-(\theta/\theta_c)^2}}{\Gamma(m/2 + 1)}. \quad (1.37)$$

The fundamental order is reduced by

$$\frac{A_0}{A_{00}} = \exp \left((-\theta/\theta_c)^2 \right) \quad (1.38)$$

1.7 Concluding remarks

Chapter 2

Balanced Homodyne for LIGO A+

There doesn't need to be a discussion about what BHD is, but it is worth briefly mentioning what it is and the other readout methods again Context for BHD upgrade being part of A+.

2.1 BHD versus DC readout for A+

- Reduction of SRC length coupling. Should this be a quantitative analysis or shall I quote the results?
- Reduction of Back scatter from the septum window between ham5 and ham6. Technically this is not to do with BHD, this could, but practically it kind of is?
- Ability to tune homodyne phase is cool but in the context of GW detection, maybe not important.
- PD SNR is not important.

- There is an optomechanical coupling between SRC cavity length noise and GW signal channel, but as far as I can tell this is well below the noise floor.

some useful references:

- G2000962, DDR CDR presentation A.
- G2001026, DDR CDR presentation B.
- Ken's notes on BHD benefits compared to DC readout, email on 25/06/2020 11:26
- LIGO-L2000200 Q and A. Useful technical points and things people will find confusing/may ask about.

Main benefit of BHD is reduction of SRC length noise coupling to AS port. This noise coupling is described in [1], see equation 13. This paper has some coupled cavity maths in it. If this needs explained, look up Transfer Functions for Fields in a 3-Mirror Nested Cavity, [2]. There is an opto-mechanical coupling between the length of the SRC and arm cavity (due to radiation pressure). Reducing the DC offset reduces the radiation pressure, which reduces this coupling.

Estimate reduction of SRC length noise coupling Differential DC offset is ϵ , so this coupling scales with the offset. Since we get the LO from elsewhere, the amount of offset can be reduced. Currently it is TEM00 is decreased to fringe offset (rms + DC) + 1mW of common mode carrier, currently there is 20mA of photo-current. (shelia dwyer answer in email FW: BHD review committee request & document PREVIEW). Assume 0.85 A/W ($q_e = 99\%$), this gives 23mW of light.

This can be reduced to between 1mW to 5mW, giving a reduction of at most 23, at least 4.6. Need to check this point.

Optical spring effect which is porbably not important In a large scale gravitational wave detector, the main advantage of BHD versus DC-readout is the ability to reduce the differential arm offset that is required to provide the GW signal a local oscillator field to beat with, as the LO field is provided by other means. SRC length noise scales quadratically with the DC offset [1], so a reduction in the DC offset causes a reduction in this coupling. SRC length noise couples to the output signal via radiation pressure. The change in differential arm length due to a change in length of the SRC scales linearly with the DC offset ($\epsilon = kL_{\text{offset}}$) due to radiation pressure. Don't know what the current DC offset is. The optomechanical coupling between SRC length and differential arm length is given as

$$\frac{\partial L_-}{\partial l_s} = \frac{16g_s^2 r_a' r_s}{ct_s^2 s_\mu^2 (1 + s_{\text{rse}})} \epsilon k P_a \quad (2.1)$$

and the normalised response between the power at the AS port and the SRC length noise is

$$\frac{S_{\text{as}}}{S_{\text{dc}}} = 4g_p^2 g_s^2 r_a'^2 \epsilon \left(\frac{\partial L_-}{\partial l_s} \right) \frac{k \Delta l_s}{1 + s_{\text{rse}}} \quad (2.2)$$

2.2 Local Oscillator Path Length Stability Requirement and HRTS requirement

The requirement for the path length stability of the local oscillator is determined by the sensitivity goal for A+. Deconvolving the strain sensitivity curve for A+ with the linear response function between differential strain, h , and power at the AS port gives the noise of the AS beam. This is then converted to an equivalent phase noise between the AS the LO beams. Finally, the requirement for the suspended optics is derived.

A gravitational wave produces a phase modulated signal in the AS beam. The LO beam will also be phase modulated by the motion of the mirrors in the LO path. When these beams are combined, it is the phase difference between the two fields that produces a signal, so the phase noise of the LO path must be sufficiently low that the signal that is generated by the interference of the LO and AS beams corresponds to the phase modulation of the AS beam.

2.2.1 Local Oscillator Path

Two options for the local oscillator path were considered: a beam picked off from the main beam splitters AR coating (BSAR) or the beam transmitted through PR2 [3] (POP). The determining factor in choice of LO path was the effect of backscatter, and the POP choice is far superior to the BSAR choice at all frequencies and scatter phases. This is because any backscatter from the BSAR enters only one arm, and therefore is a differential effect. Any back scatter that enters the POP port is common-mode, meaning that the differential arm signal is far less sensitive to this

source of backscatter.

The layout for the optics which combine the the LO and AS beam is shown in figure 2.1. The beam transmitted by PR2 is used for control of the interferometer, so BHDBS1 is needed to pick off some of this light. The LO beam is steered towards the combining beam splitter BHDBS2. with BHDM1. Two lenses (**need names for these**) are used for mode matching the LO beam to the AS beam. The AS beam is transmitted through the SRM and OFI, and then is steered towards BHDBS2 with OM0.

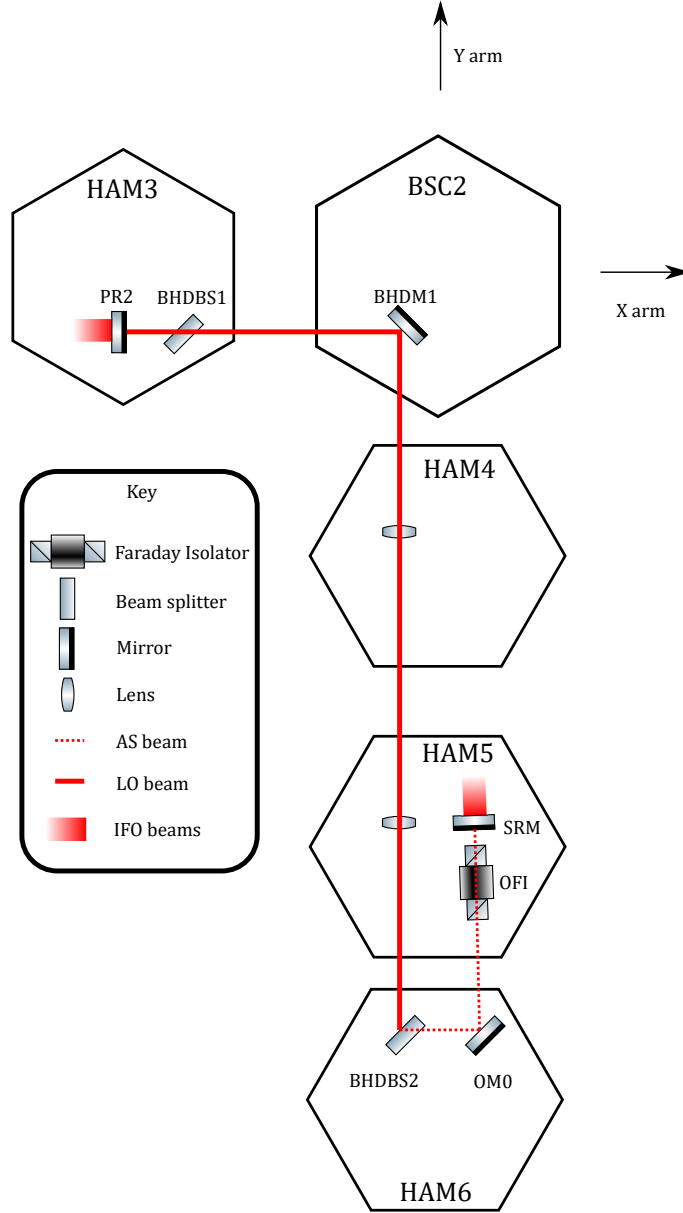


Figure 2.1: Sketch of the LO and AS beam paths. The LO beam is obtained from the light transmitted by PR2. Some of this light is needed for control of the interferometer, so BHDBS1 is used to split the light. The LO beam is then directed towards the combining beam splitter BHDBS2 with BHDM1. Two lenses are needed to mode match the LO beam to the AS beam. The AS beam is transmitted through the SRM, passes through the OFI, and is directed to BHDBS2 via OM0.

2.2.2 Simulation of the Linear Response Function Between a Differential Strain and the Power at the AS Port

The linear response function of the A+ detectors was obtained using FINESSE. Since the core optics of the interferometer will not be changing in a way relevant to this calculation, the LLO model[4] was used to calculate the response between differential motion of the ETMs and the power transmitted through the OMC. To apply a differential arm signal with FINESSE, `fsig` commands were used to apply two signals in antiphase with each other to the ETMs,

```
fsig sig1 ETMXHR 1 0
fsig sig2 ETMYHR 1 180.
```

The frequency of this modulation was swept, and the signal produced at the AS port was measured by demodulating the light at the modulation frequency. The commands

```
xaxis sig1 f log 5 5000 2999
put P_DC_OMC_probe f1 $x1
```

do this. Since the input modulation is doubled by moving the mirrors differentially the response needs dividing by 2. To account for the ETMs retro-reflecting the light (or is the weird Finesse convention), the response needs dividing by 2 again. To convert this from phase units to strain units, a factor of $\frac{\lambda}{2\pi L_{\text{arm}}}$ is needed. Therefore, the output of the the FINESSE simulation needs to be multiplied by $\frac{1}{2} \frac{1}{2} \frac{2\pi L_{\text{arm}}}{\lambda}$ to be in units of $\frac{\text{W}}{\text{h}}$. The linear response function between a differential arm strain and the power at the AS port is shown in figure 2.2.

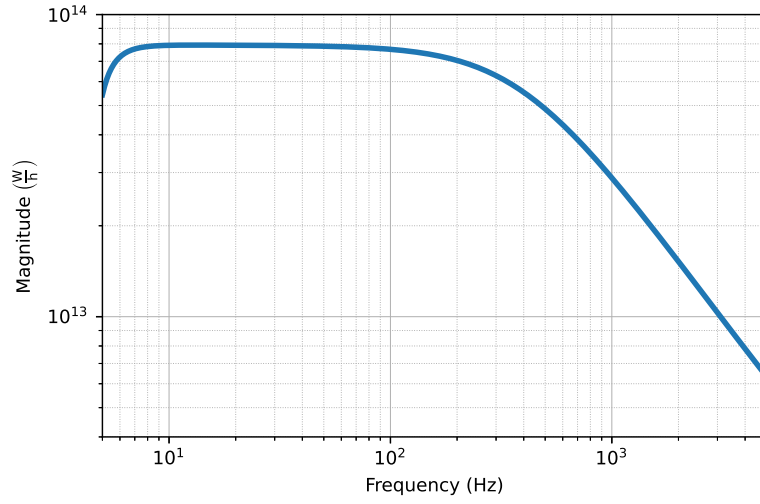


Figure 2.2: The linear response function between strain and light power at the AS port was obtained from a FINESSE simulation of the LLO detector[4]. The corner frequency at ≈ 300 Hz comes from the pole of the coupled SRC-arm cavity. The behaviour below 6 Hz is due to optical spring effects.

2.2.3 Converting the Sensitivity Curve for A+ into the HRTS requirement

The sensitivity curve will be shown in an earlier chapter. To convert the A+ sensitivity curve into a path length stability requirement for the local oscillator, a few steps are required. First, the sensitivity curve must be expressed in units of $\frac{W}{\sqrt{\text{Hz}}}$. Secondly, the intensity noise of the AS beam needs to be converted into a phase noise. In order for the phase noise of the local oscillator to not limit the performance of the gravitational wave detector, the phase noise of the AS beam must be greater than the phase noise of the LO beam. Finally, a factor of 10 safety margin is required. The HRTS requirement is then obtained by including the number of reflective optics in the LO path.

The A+ sensitivity[5] curve was multiplied by the function shown in 2.2 to convert it from units of $\frac{h}{\sqrt{\text{Hz}}}$ to $\frac{W}{\sqrt{\text{Hz}}}$. This is shown in figure 2.3. At frequencies above 1 kHz, the requirement is flat as this is the region dominated by shot noise. At lower frequencies, the requirement is relaxed as the sensitivity curve becomes dominated by other noise sources.

The quantum mechanics of this problem have been included in the A+ sensitivity curve [5], therefore we can approach this with classical physics. When the LO and AS beams interfere at the beam splitter (BHSBS2), the power of the resulting beam is

$$P = |E_s e^{i\omega t} + E_{\text{LO}} e^{i\omega t + \phi}|^2 = |E_s|^2 + |E_{\text{LO}}|^2 + 2|E_s E_{\text{LO}}^\dagger| \cos(\phi), \quad (2.3)$$

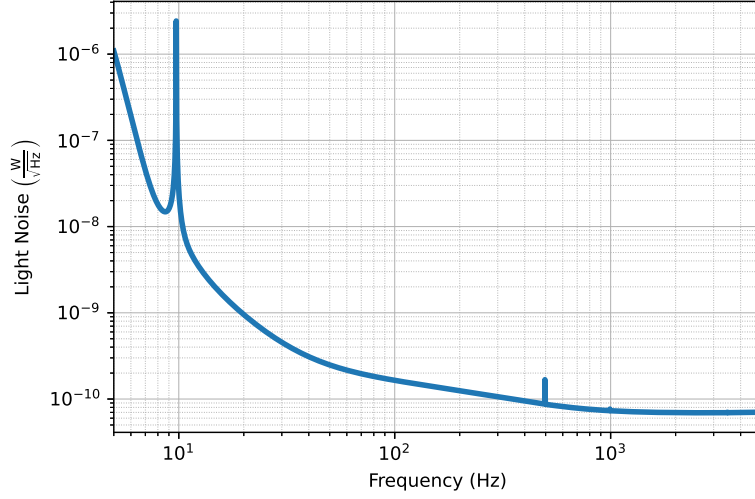


Figure 2.3: The intensity noise of the light at the AS port for the A+ detector. This was obtained by deconvolving the detectors sensitivity curve[5] with the function shown in figure 2.2.

where E_s and E_{LO} are the (complex) field amplitude of the signal and local oscillator beams, ω is the laser frequency, t is time, and ϕ is the phase difference between the local oscillator and the signal beam, also known as the homodyne angle. Differentiating with respect to ϕ gives the relationship between a change in power and a change in phase between the two beams,

$$dP = -2|E_s E_{LO}^\dagger| \sin(\phi) d\phi. \quad (2.4)$$

and since the homodyne angle is 90° , this becomes

$$dP = -2|E_s E_{LO}^\dagger| d\phi. \quad (2.5)$$

To convert ϕ to distance, use

$$\frac{\lambda}{2\pi}d\phi = dx. \quad (2.6)$$

The light noise curve shown in figure 2.3 should be multiplied by $\frac{\lambda}{2\pi} \frac{1}{2|E_s E_{\text{sig}}^*|}$ to obtain the the noise floor of the interferometer in units of LO path length noise. For a 98 mW LO beam and a 5 mW AS beam, the noise floor of the interferometer in units of LO path length is shown in figure 2.5.

The HRTS requirement assumes, based on the current design, that there will be 2 mirrors at 45° which are used to steer the AS and LO beams. As the optics reflected at 45 degrees, the path length change due to the optics motion is multiplied by $\frac{2}{\sqrt{2}} = \sqrt{2}$. The beams are combined on a beamsplitter, and the notion of this beam splitter needs to be counted twice. A sketch of this is shown in figure 2.4. A factor of 10 is then included as a safety margin since there are many uncorrelated noise sources in LIGO. With a 98 mW LO beam and a 5 mW AS beam, the requirement for the HRTS optics longitudinal motion at 100 Hz is $5 \times 10^{-17} \text{m}/\sqrt{\text{Hz}}$. A typical motion of the optical benches at 10 Hz is $10 \times 10^{-1} \text{m}/\sqrt{\text{Hz}}$ [6]. To achieve this with passive isolation, a triple suspensions will be used. make figure showing requirement as a fuction of AS power?

2.2.4 HRTS

remove references to HRTS in previous section. Refer to the HRTS there as table motion isolation requirements, or something like that. Discuss the HRTS in its own separate section.

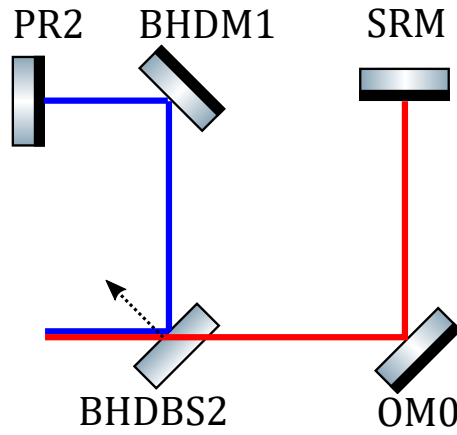


Figure 2.4: A sketch of the LO and AS beam paths. The LO beam path is shown in blue and the AS beam is shown in red. A dashed line indicates the direction that the recombining beamsplitter, BHDBS2, moves. When this happens, at the point of the beams combining the phase of the AS beam has increased while the phase of the LO beam has decreased. This means that the recombination beamsplitter motion needs to be counted twice.

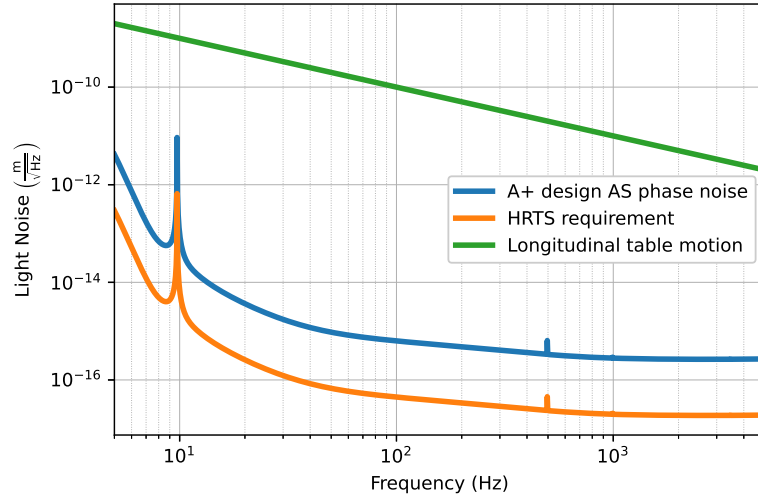


Figure 2.5: The phase noise between the LO and the AS beams causes noise. The sensitivity curve for A+ sets the limit on how much path length noise is acceptable between the AS and LO beams. The blue line shows sensitivity curve is shown in units of path length noise between the AS and LO beams, and the orange line shows the requirement of HRTS mirror motion.

2.3 Alignment of the LO and AS beam

2.4 HAM6 layout

Fundamentally, the layout in HAM6 must combine the AS beam and the LO beam, and sense the combined beams to form a balanced homodyne detector. There are several other functionalities required and constraints imposed on the layout of HAM6. The following section will describe these functionalities and constraints and how they guided the design of the HAM6 layout. This work was undertaken by a team including myself, Ken Strain, Russel Jones, Stephen Webster and Mark Barton.

2.4.1 The Constraints and Requirements of the Layout of HAM6

The entry point of the AS and LO beams in principle can be anywhere in HAM6, however in reality they are fixed due to the location of HAM6 with respect to the interferometer, due to the location of the septum windows between HAM5 and HAM6, and due to the many other components of the interferometer that these beams need to avoid. The LO beam is steered with BHDM1 onto BHDBS2, and the AS beam is steered with OM0 onto BHDBS2. This is required to provide the degrees of freedom to co-align the AS and LO beams. There is some freedom in choice for the location of OM0 and BHDBS2, but there is no choice in the location of the AS and LO beams.

The design beam parameter at the SRM is $q_{\text{SRM}} = -3.55 + 1.09i$ m, so the design of the HAM6 mode matching optics is based on this number. In reality, the beam parameter could be different to this, as discussed in chapter ??). At the SRM, there are two sets of modes we could design the AWC to match to: The arm mode, or the SRC mode. Ideally, these would be the same mode, however geometry errors in the SRC cause the arm and the SRC to be mismatched meaning the radius of curvature and beam size at the SRM will not correspond to the eigenmode of the SRM. This set of likely modes, shown in ??, was taken into consideration when choosing a layout for the AWC in HAM6.

After BHDBS2, the layout the optics between OMC input couplers and BHDBS2 must provide the functionality for steering the combined AS and LO beam into the OMC, and provide mode matching for the AS beam to the OMC. In addition to this, sensors in HAM6 are required for targeting the AS and LO beams. Some of these functions will be achieved using beams which are picked off from the main beam path, and so additional mirrors will be required to steer these beams around the chamber. A set of smaller optics will be installed to provide diagnostic beams which can be imaged with CCDs. A shutter will be required to protect the hardware in HAM6 from being damaged by the large amounts of light that can potential be transmitted through the SRM.

After the AS and LO beams are combined, double stage suspensions will guide the beam from BHDBS2 to the OMC. These suspensions provide active positioning control and active wave front control. In principle, the role of the AWC in HAM6 is to correct for geometry errors of the optics after the SRM such that the mode

matching between the Arm mode and the OMC waist is maximised. This assumes that the changes in the interferometers mode, for instance due to a change in thermal state, will be corrected for by other subsystems e.g. the TCS. In practice, operators of the interferometer do what ever improves the SNR of the the gravitational wave signal. For instance, it has been found that TCS can be used to decrease laser intensity noise coupling to the the GW signal channel **CHECK**. The AWC in HAM6 was designed with this flexibility in mind. The losses due to components in HAM6 is summarised in table 2.1 (See E1900385).

The goal for mode matching between the arm mode and the OMC mode is is $> 98\%$. The SAMS actuators will provide ± 100 , mD of optical power tuning range, with a bandwidth of the order 100 Hz. Astigmatism can be considered as a loss, and none-zero angles of incidence (AOI) of beams onto curved optics results in a the beam becoming astigmatic. This prevents designs using large AOI. This is described in detail in Chapter ??.

The current OMC design (with trivial modifications) will be used to remove junk light from the signal beam, so their is no choice in target waist size. The input beam at the SRM is defined in T1200410. The mode matching is designed assuming that the beam at the SRM is the ‘design mode’. Mode matching between the beam exiting the SRM and the OMC waist will be done with a static lens and 2 active mirrors. The static lens will be placed just after the OFI and its purpose is to provide sufficient Gouy phase separation of the 2 active mirrors.

The light transmitted OMCs is used to measure the gravitational wave signal, since the higher order modes are suppressed by a factor of the finesse of the OMC.

The OMCs are expected to be the largest contributor of back-scatter from HAM6 back to the interferometer. By having the OMCs after the AS and LO beams are combined, and by having the OFI protect the interferometer, the requirement on the motion of the OMCs is fairly lax. Nevertheless, a single stage suspended platform is required. This will provide isolation down to 4 Hz.

HAM6 loss Source	Target for O5
OMC throughput	< 2%
OM3 transmission (or equivalent) for sensing	1–2%
PD quantum efficiency	1%
Mode mismatch to the OMC	< 2% (as low as possible)
Any other optical loss	< 1%

Table 2.1: Target for losses in HAM6. Reproduction of E1900385

Space Name	Optical Path Length (mm)
SRM to OFI lens	1120
OFI lens to HAM6 table edge	720
HAM6 table edge to OM0	432
OM0 to BHDBS2	905
BHDBS2 to OMxS	225
OMxS to OMx1	1240
OMx1 to OMx2	1240
OMx2 to OMCx waist	1240

Table 2.2: Spacings for the main beam path from the SRM to the OMC waist.

2.4.2 Mode Matching and Control

Hermite Gaussian beams will be explained earlier in the intro chapters, so don't need generic expansion, just use the 2nd order ones. Have a read of [10].

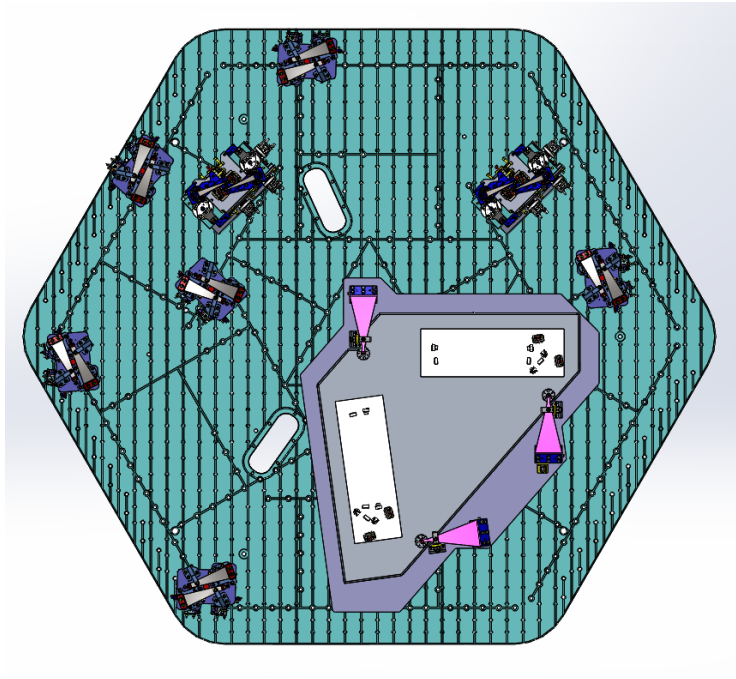


Figure 2.6: Place holder graphic. The pentagram layout. CAD will be inserted later.

The mode-matching between two laser beam modes, or a laser mode and a cavity mode, describes how similar those two modes are. In general, the two modes can be mismatched by being misaligned or by having a different beam shape. Mode matching typically refers to the overlap of the two modes if they were perfectly aligned.

Mode-matching is usually, to first approximation, thought of as a difference in beam waist size and beam size location. This can be mathematically described by the $\text{HG}_{20} + \text{HG}_{02}$ mode. The effect of changing the waist location of the mode is equivalent to an in phase change in HG_{20} content, whereas an change in waist location is orthogonal to this. [read up on this](#). Sensing the $\text{HG}_{20} + \text{HG}_{02}$ mode therefore gives a measure of the mode-matching between two lasers. As the $\text{HG}_{20} + \text{HG}_{02}$ acquires 3ϕ compared to a plane wave, and 2ϕ compared to the HG_{00} mode, spacing the actuators at 45° will give orthogonal control of the mode matching.

To correct for mode-mismatch, the optical layout will be such that the AWC can orthogonally actuate on the beam parameter of the arm mode when its reached the OMC.

The two mode will not perfectly interfere, and this results in some signal loss. The amount of mode matching $\mathcal{M}$ between two astigmatic modes modes, $q_{1,2}$, is

$$\mathcal{M} = \frac{\sqrt{(q_{x1} - q_{x1}^*) (q_{y1} - q_{y1}^*) (q_{x2} - q_{x2}^*) (q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|}. \quad (2.7)$$

As well as mode matching, the layout in HAM6 should provide orthogonal control for alignment. As the alignment can be described by the $\text{HG}_{10} + \text{HG}_{01}$, the alignment

Optic Name	Distance along optical axis (m)
SRM	0.000
OFI lens	1.122
OM0	2.272
BHDBS2	3.177
OMxS	3.402
OMx1	4.642
OMx2	6.232
OMC Waist	7.452

mode acquires an extra ϕ of Gouy phase compared to the HG_{00} mode. Orthogonal control can be achieved by separating the actuators for this by 90° .

2.4.3 Chosen Layout

Optically, there are many layouts which could have satisfied the above requirements and constraints. It was decided that the two beams after BHDBS2 would have the same path length to reduce the cost of the HAM6 optics. Ultimately the choice of layout for HAM6 was determined by mechanical factors such as table space and number of allowable suspensions. The design that is being taken forward at the time of writing is known as the ‘pentagram’, and is shown in figure 2.6.

Show plots produced by `ligo_layout.py`

2.4.4 WS visualisations as a tool for understanding mode matching problems

[11]

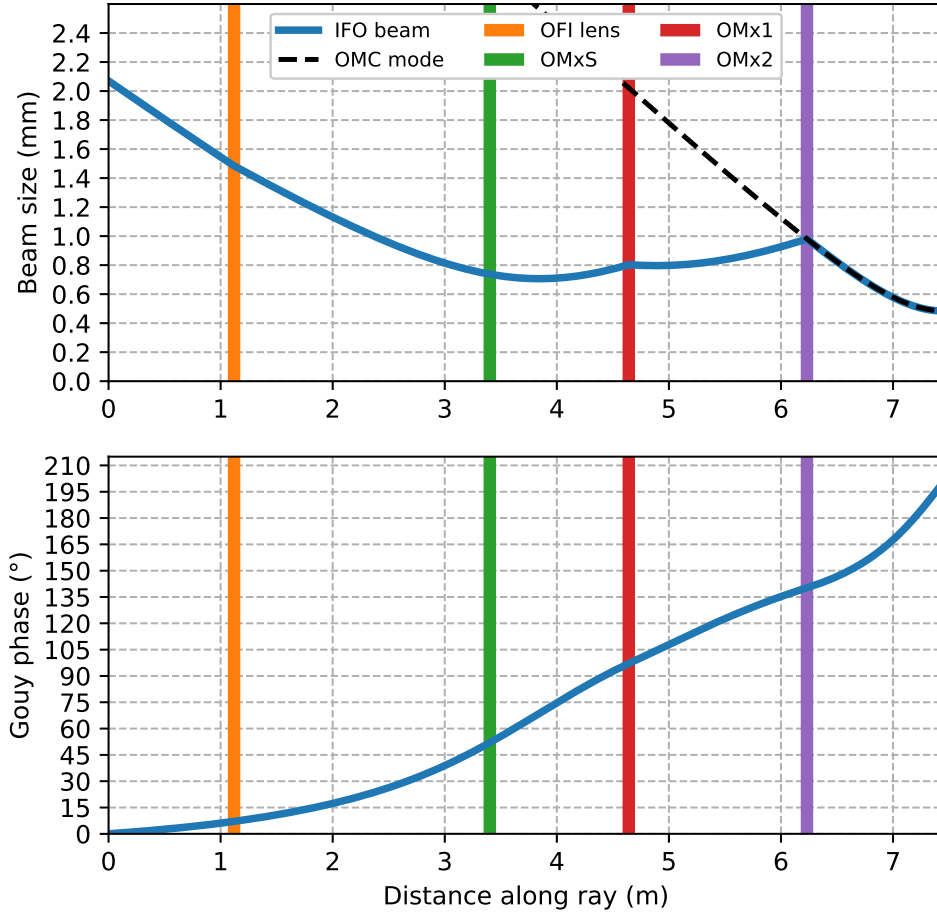


Figure 2.7: The radius of the beam as it travels from the SRM ($z = 0$) to the waist of the OMC.

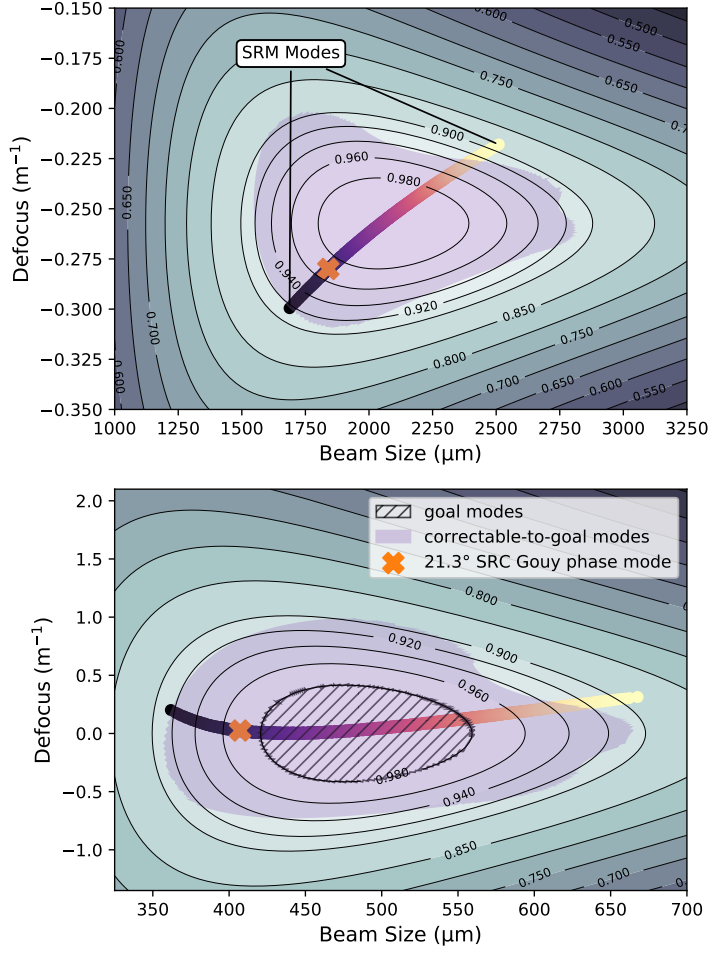


Figure 2.8: Top panel: the set of modes that could be present at the SRM is shown by the black-purple-orange-white curve, as described in chapter ???. The purple region shows which modes can reach 98% mode matching by using the the AWC in HAM6. The orange cross indicates the mode that corresponds to the SRC having a single pass Gouy phase of 21.3°. The contours indicate the mode matching between the arm mode and the OMC mode.

Bottom panel: this shows the same modes as the top panel, except they have been transformed to the plane of the OMC waist. The target region is highlighted with a hatched area.

2.5 Conclusion about BHD for $A+$

Appendix A

Mode Matching Between 2 Astigmatic Beams

A.1 Derivation of Mode Matching TM00 Beams

The [check this naming](#) electric field, E , of a TM00 Gaussian beam can be written as

$$E(x, y, z) = E_x(x, z)E_y(y, z) \quad (\text{A.1})$$

where

$$E_x(x, z) = \frac{1}{\sqrt{q_x(z)}} \exp\left(-ik \frac{x^2}{2q_x(z)}\right), \quad (\text{A.2})$$

$$E_y(y, z) = \frac{1}{\sqrt{q_y(z)}} \exp\left(-ik \frac{y^2}{2q_y(z)}\right). \quad (\text{A.3})$$

For a beam without astigmatism, $x^2 + y^2 = r^2$ and $q_x = q_y = q$ so:

$$E(r, z) = \frac{1}{q(z)} \exp\left(-ik \frac{r^2}{2q(z)}\right). \quad (\text{A.4})$$

The mode matching, $\mathcal{M}$, of two beams can be expressed by overlap of the two fields

$$\mathcal{M} = \frac{\left| \int_A E_1^* E_2 dA \right|}{\int_A |E_1|^2 dA \int_A |E_2|^2 dA}. \quad (\text{A.5})$$

The numerator of equation A.5 for TM00 beams can be written as:

$$\left| \int_A E_1^* E_2 dA \right| = \int_A E_1^* E_2 dA \int_A E_1 E_2^* dA \quad (\text{A.6})$$

$$= \int_{-\infty}^{\infty} E_{x1}^* E_{x2} dx \int_{-\infty}^{\infty} E_{y1}^* E_{y2} dy \int_{-\infty}^{\infty} E_{x1} E_{x2}^* dx \int_{-\infty}^{\infty} E_{y1} E_{y2}^* dy. \quad (\text{A.7})$$

Since the calculation is the same for the x and y coordinates, we only need to show the solution for the x . The x terms of equation A.7 can be written as:

$$\int_{-\infty}^{\infty} E_{x1}^* E_{x2} dx \int_{-\infty}^{\infty} E_{x1} E_{x2}^* dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{q_{x1}^*}} \exp\left(\frac{ikx^2}{2q_{x1}^*}\right) \frac{1}{\sqrt{q_{x2}}} \exp\left(\frac{-ikx^2}{2q_{x2}}\right) dx \quad (\text{A.8})$$

$$\times \int_{-\infty}^{\infty} \frac{1}{\sqrt{q_{x1}}} \exp\left(\frac{-ikx^2}{2q_{x1}}\right) \frac{1}{\sqrt{q_{x2}^*}} \exp\left(\frac{ikx^2}{2q_{x2}^*}\right) dx \quad (\text{A.9})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2}{2q_{x1}^*}\right) \exp\left(\frac{-ikx^2}{2q_{x2}^*}\right) dx \times c.c. \right) \quad (\text{A.10})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2}{2q_{x1}^*} - \frac{-ikx^2}{2q_{x2}^*}\right) dx \times c.c. \right) \quad (\text{A.11})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2 (q_{x2} - q_{x1}^*)}{2q_{x1}^* q_{x2}}\right) dx \times c.c. \right). \quad (\text{A.12})$$

Using the identity

$$\mathcal{I} = \int_{-\infty}^{\infty} \exp(iax^2) dx = \frac{i\pi}{a} \quad (\text{A.13})$$

and labelling

$$a = \frac{k(q_{x2} - q_{x1}^*)}{2q_{x1}^*q_{x2}} \quad (\text{A.14})$$

allows equation A.12 to be simplified to

$$\frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2(q_{x2} - q_{x1}^*)}{2q_{x1}^*q_{x2}}\right) dx \times c.c. \right) = \frac{1}{|q_{x1}| |q_{x2}|} \sqrt{\frac{i\pi(2q_{x1}^*q_{x2})}{k(q_{x2} - q_{x1}^*)}} \sqrt{\frac{-i\pi(2q_{x1}q_{x2}^*)}{k(q_{x2}^* - q_{x1})}} \quad (\text{A.15})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \sqrt{\frac{4\pi |q_{x1}|^2 |q_{x2}|^2}{k^2 |q_{x2} - q_{x1}^*|^2}} \quad (\text{A.16})$$

$$= \frac{2\pi}{k} \frac{1}{|q_{x2} - q_{x1}^*|}. \quad (\text{A.17})$$

This means that the numerator of equation A.5 is

$$\frac{4\pi^2}{k^2} \frac{1}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|}. \quad (\text{A.18})$$

Since E_x and E_y are orthogonal, $|E|^2 = |E_x|^2 |E_y|^2$, an expression for the denominator of equation A.5 can be found by:

$$\int_A |E|^2 dA = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |E_x|^2 |E_y|^2 dx dy \quad (\text{A.19})$$

$$= \int_{-\infty}^{\infty} \frac{1}{|q_x|} \exp\left(\frac{ikx^2(q_x - q_x^*)}{2q_x q_x^*}\right) \times y \text{ term.} \quad (\text{A.20})$$

Using a similar approach as in the calculation of the numerator terms:

$$\int_A |E|^2 dA = \frac{1}{|q_x|} \sqrt{\frac{i2\pi |q_x|^2}{k(q_x - q_x^*)}} \frac{1}{|q_y|} \sqrt{\frac{i2\pi |q_y|^2}{k(q_y - q_y^*)}} \quad (\text{A.21})$$

$$= -\frac{2\pi}{k} \frac{1}{\sqrt{(q_x - q_x^*)(q_y - q_y^*)}} \quad (\text{A.22})$$

we get the mode matching $\mathcal{M}$ to be

$$\mathcal{M} = \frac{\frac{4\pi}{k^2}}{-\frac{2\pi}{k} \times -\frac{2\pi}{k}} \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)} \sqrt{(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|} \quad (\text{A.23})$$

$$= \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|}. \quad (\text{A.24})$$

A.2 Effect of Error in the Radius of Curvature of a Mirror on the defocus of a Beam

The error in the optical power of a mirror due to an error in the radius of curvature of is

$$S = \frac{2}{R_m} \quad (\text{A.25})$$

$$dS = \frac{-2}{R_m^2} dR. \quad (\text{A.26})$$

The error in the beams defocus is the same as the error in the radius of curvature of the the mirror. The effect this has on the beam parameter q can be found by

defining a contour of modes distance dq from q i.e. $q + |dq| e^{i\theta}$, where dq is

$$q = \frac{q_i}{Sq_i + 1} \quad (\text{A.27})$$

$$dq = \frac{q_i^2}{(-Sq_i + 1)^2} dS. \quad (\text{A.28})$$

This gives you a family of modes which will contain the modes due to an error in the radius of curvature of the mirror, but since q is a complex number the contour you need to define to encapsulate the concept of an error in q also contains modes which are have different beam sizes as well as different defocuses. To do the calculation this way is cumbersome, it is much more straight forward to compute the effect of having a mirror ROC error by evaluating at every mirror

$$\begin{pmatrix} 1 & 0 \\ \pm dS & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -S & 1 \end{pmatrix} \begin{pmatrix} q \\ 1 \end{pmatrix} = \begin{pmatrix} q \\ (S \pm dS)q + 1 \end{pmatrix} \quad (\text{A.29})$$

and then divide the top and bottom entries of the $\begin{pmatrix} q \\ (S \pm dS)q + 1 \end{pmatrix}$ vector.

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